

MEDICINE PRESCRIPTION AUTOMATION

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**A project report submitted in fulfilment of the
Requirements for the award of the degree of
BS Software Engineering**

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18 FEB 2022

DECLARATION

We hereby declare that this project report is based on our original work except for citations and quotations which have been duly acknowledged. We also declare that it has not been previously and concurrently submitted for any other degree or award at Karachi University or other institutions.

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MEDICINE PRESCRIPTION AUTOMATION

ABSTRACT

Doctor's prescription is basically the text which is hand-based written by doctors' as recommendations which include lists of medicines related to health issues, injuries & sickness problems. On getting prescription from doctor mostly a normal person is not able to understand what the prescribed medicine is. "In most cases, however, we wouldn't be able to read it anyway because doctors use Latin abbreviations and medical terminologies on prescriptions that are not understandable by the general persons which make reading it very difficult [1]".

This project will resolve the problem in Doctor's prescription through AI and Neural Networks to introduce this software. The reason behind using this technique is that it enable with the qualities of providing features-extraction & feature-detection which is the most reliable for Optical Character Recognition. After many attempts and bugs, an appropriate set of trained parameter are specified and network structure which includes number of layers like output layer, input layer, and hidden layer and accordingly neurons are created.

The very step that takes place is pre-processing of taken captured picture along with its inverting & threshold followed by smoothing. And then filtering, resizing, segmentation & feature extraction being followed. "Feature extraction primarily refers to the extraction of each character from the image [2]". Next, output matrix is given in by invoking the network. The character then can be determined depending on this output matrix. This system is being made to automate the network for any human.

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LIST OF SYMBOLS / ABBREVIATIONS

AI	Artificial Intelligence
NN	Neural Network
CNN	Convolutional Neural Network
RELU	Rectified Linear Unit
IPO	Input-Process-Output
RGB	Red Green Blue
OCR	Optical Character Recognition
LSTM	Long Short-Term Memory
API	Application programming interface
GUI	Graphical User Interface
RPC	Remote Procedure Call
RST	Representational State Transfer
ML	Machine Learning
Tab	Tablet
Cap	Capsule
Inj	Injection
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
URL	Uniform Resource Locator

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CHAPTER 1

INTRODUCTION

1.1 Background

Medicine prescription is handwritten document by a doctor in which the doctor writes the medicines name and medicines intake time patterns. Most of the time, a common man is not able to read the doctor's handwriting, which can mislead them on medicine intake time and quantity. We will be working on text recognition of doctor's handwriting and decode it in computerized digital format which is very easily readable.

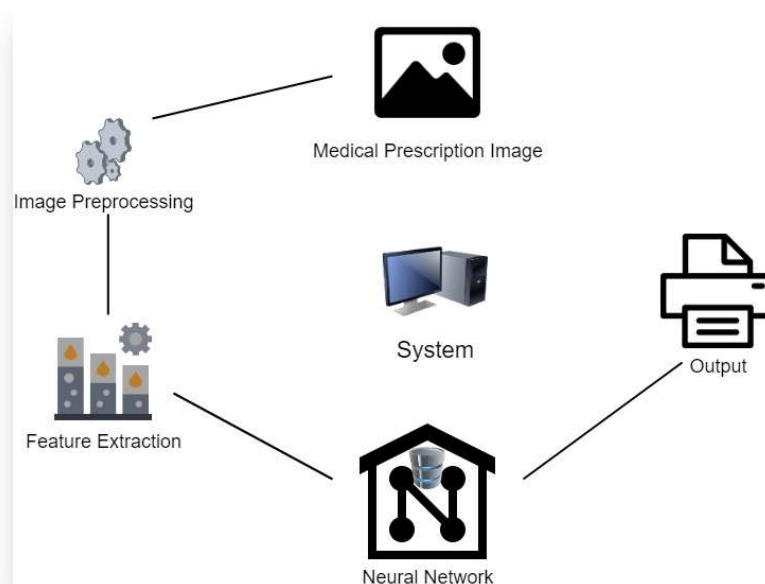


Figure 1.1: System Structure

1.2 Problem Statements

This application is based on handwritten text recognition. Currently you can easily find text recognizing applications. “Text recognition in images is an active research area which attempts to develop a computer application with the ability to automatically read the text from images [3]”. But that works only on simple & clean writings mostly. We are taking this a step further and will make it recognize doctor’s writing as well. “As Medication Error, is the 3rd leading death cause after heart attacks & cancer [4]”. “Another study says that ‘Doctors’ sloppy handwriting kills more than 7000 people annually [5]”. So we think this is useful for our community and they will then be able to know that what medicines they need to get. This will provide easiness for the people who are purchasing or selling medicines and are using medical prescriptions provided by the doctors.

1.3 Aims and Objectives

The objectives of the thesis are shown as following:

- i) Collection of general physicians prescribed medicines.
- ii) Creating our own dataset of medicines which are generally prescribed in Pakistan.
- iii) Image Processing and noise reduction and feature extraction.
- iv) Train and test a model for learning the handwriting of doctor and recognition of new prescription.
- v) To make it easy for a common man to understand the doctor’s prescription.
- vi) To provide efficient automated prescription’s in digital form.

1.4 Scope of Project

The scope of this project is to pre-process the image, extract features, apply algorithm of deep learning to recognize the medicine prescription.

Normally, as we see prescriptions are only understood by either doctors or pharmacist. Obviously we the ordinary people are not able to understand the

Drug	Brands
------	--------

mentioned medicines. Therefore, in order to overcome this critical hurdle & letting the normal people understand prescriptions by recognizing doctor handwriting and further also getting hands on medicines in Deep Machine Learning.

The data we have collected through medicine prescription which is generally prescribed in Pakistan. The structure of prescription consists of hospital name, doctor name, date, and medicine or formula name with dose. Each prescription has average of 4 medicines.

The most general medicines prescribed by the doctors in the prescriptions have been divided in to 12 classes. The dataset for training of algorithm will be based mostly on these general medicines categories.

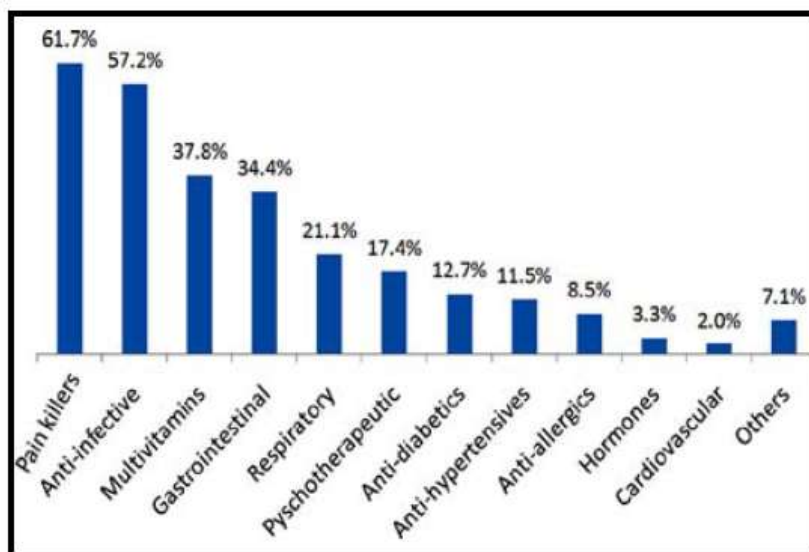


Figure 1.2: General Medicines Classes Prescribed by the Doctors

Aspirin	• Tab. Aspirin 300 mg • Tab. Ascard 75,150,300mg • Tab. Disprin 300 mg
Ibuprofen	• Tab. Brufen 200,400,600 mg • Tab. Actifen 200,400,600 mg
Flurbiprofen	• Tab. Froben 50,100 mg • Tab. Ansaid 100 mg
Diclofenac Sodium	• Tab. Artifen 25,50 mg • Tab. Dicloran 50 mg • Tab. Voren 25, 50 mg
Naproxen	• Tab. Flexin 250,500 mg • Tab. Synflex 550 mg
Piroxicam	• Tab. Brexin 20 mg • Tab. Feldene 20 mg
Nimesulide	• Tab. Nims 100 mg • Tab. Unix 100 mg
Ketorolac	• Inj. Toradol 10,30 mg/ml
Paracetamol	• Tab. Panadol 500 mg • Tab. Calpol 500 mg • Tab. Disprol 500 mg
Paracetamol + caffeine	• Tab. Panadol Extra • Tab. Calpol Plus
Paracetamol + Orphenadrine	• Tab. Wilgesic (450/35 mg) • Tab. Wilgesic Forte (650/50mg) • Tab. Nuberol Forte (650/50mg)
Paracetamol + Codeine	• Tab. Codogesic (500/15 mg)
Tramaadol	• Cap. Tramal 50 mg • Inj. Tramal 50 mg/ml IM/IV
Tizanidine (Musle Relaxant)	• Ternelin 2,4 mg • Movax 2,4 mg

Table 1.1: General Medicines Classes Prescribed by the Doctors [6]

Most Common type of pain killer medicines taken from prescription and internet

Medicine Type	Prescribed %
Pain Killers	51%
Anti-allergies	10%
Anti-infective	8%
Cardio-vascular	6%
Hormones	3%
Multivitamins	10%
Others	12%

Table 1.2: Observed Ratio from our collected dataset

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Under this chapter the relevant approaches on the current issues and challenges of pictures that are based on handwritten and their recognition system. The mentioned information and theories are all gathered from books, journals, websites and etc.

Initially in this chapter we will be discussing the recognition of images which would be related to handwriting which is picture based and the recognition system of digital document.



Figure 2.1: OCR process from scanned doc, image to text

2.2 Google Vision API

Helping in understanding your images with industry major prediction accuracy, the great Google Clouds presents the two computer vision product using machine learning.

- AutoML Vision
- Vision API

2.2.1 Vision API

The great “Google Cloud’s Vision API” has introduced well-built and tough pre-trained mechanism learning models by the help of “REST” and “RPC” APIs. This helps in assigning tags or labels to images and at a great speed categorizing following into many of the categories. “Detect objects and faces, read printed and handwritten text, and build valuable metadata into your image catalogue [7]”.

- Use APIs
Use REST and RPC APIs.
- Helps in classifying the image using its present tags
The pre-trained models purchase extensive libraries of given tags.
- Google’s data labeling is also used
We can help you in explaining the given image, video and words.
- At the side, deployment of mechanism learning is made.
Deployment of lower suspension and high correctness models improved for side gadget.
- Detection of objects
Detection of objects that where they exist, and how much.
- validate innovative product searching
Comparison of picture to images in your outcome index, and then return a list of similar items.
- Detecting handwriting and printed
Using OCR to automate, identifying of languages.
- Detection of web units and pages

It finds new affairs, symbols, and closed pictures on the web.

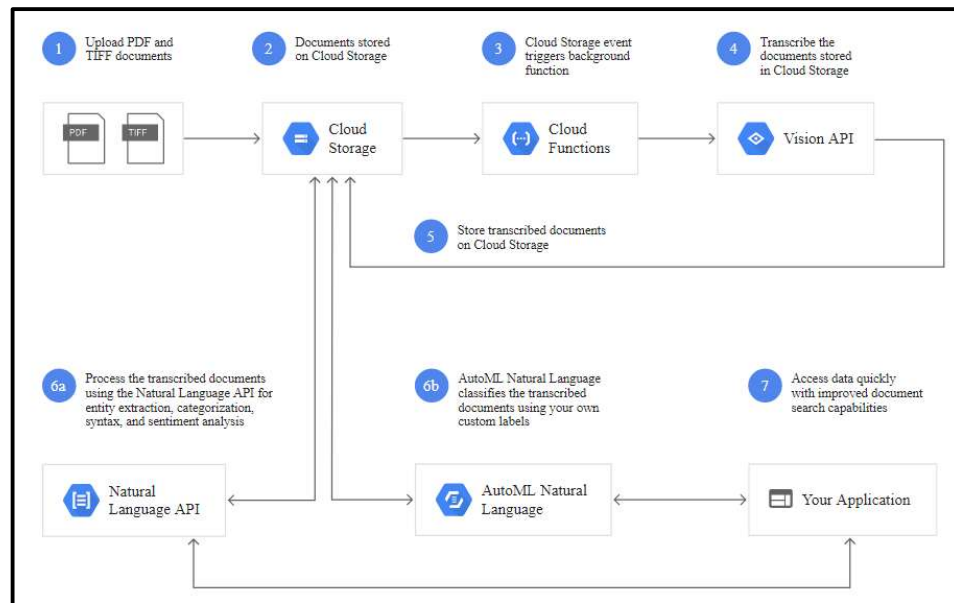


Figure 2.2: Google Vision API Document Classification Model

2.3 Tesseract – an open-source OCR

Tesseract being an “open-source OCR” mechanism, has earned great name among OCR producers. It can be hard to apply and making changes, as at that time there were not many free & well-structured “Optical Character Recognition” substitutes for so long. Then Tesseract started its “Ph.D. research project in HP Labs, Bristol” which earned demand & was introduced by “HP” during 1984-1994. In the year of 2005, the HP introduced Tesseract as free software. The year of 2006 and onwards, it was introduced by Google.

The availability of Tesseract is present under the Apache 2.0 license. It can be used in both the ways either directly, or either the use of an API to identify and detect printed-text from the taken pictures. It also supports a vast diversity of different languages. It did not inherit GUI, but various kinds are there to use from the 3rd Party pages. It is complying with number of frameworks & programming languages

with covers which are present. “It can be used with the existing layout analysis to recognize text within a large document, or it can be used in conjunction with an external text detector to recognize text from an image of a single text line [8]”.

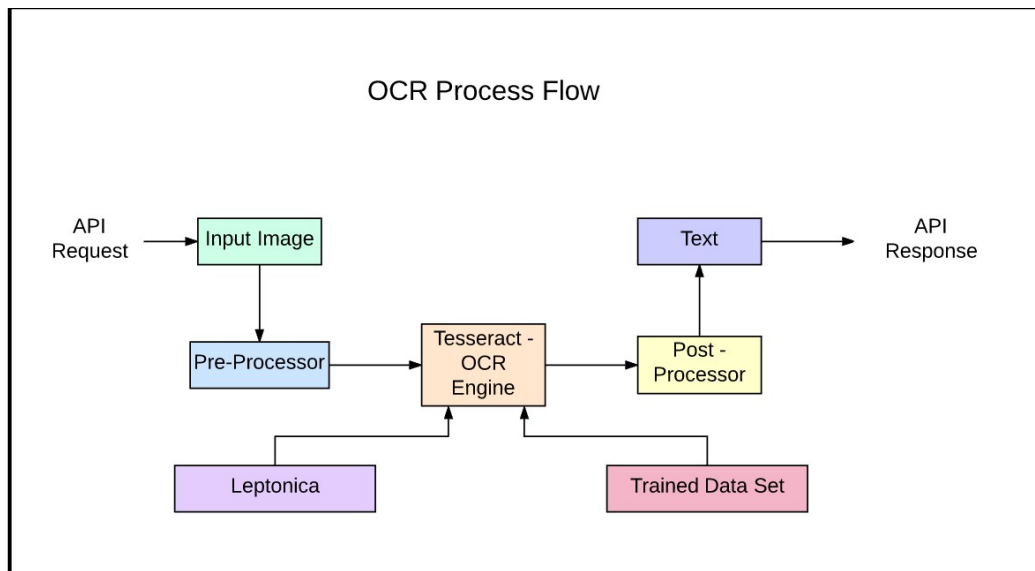


Figure 2.3: Tesseract API’s flow of process for Optical Character Recognition

2.3.1 Technology – How it works:

Tesseract was introduced from “OCRopus” prototype in Python which was a branch of LSTM in C++, known to be as “CLSTM”. In C++, CLSTM is known as the application of LSTM cyclic neural network, as well as the use of Eigen Library for solving out of the expressions that are numerical in nature.

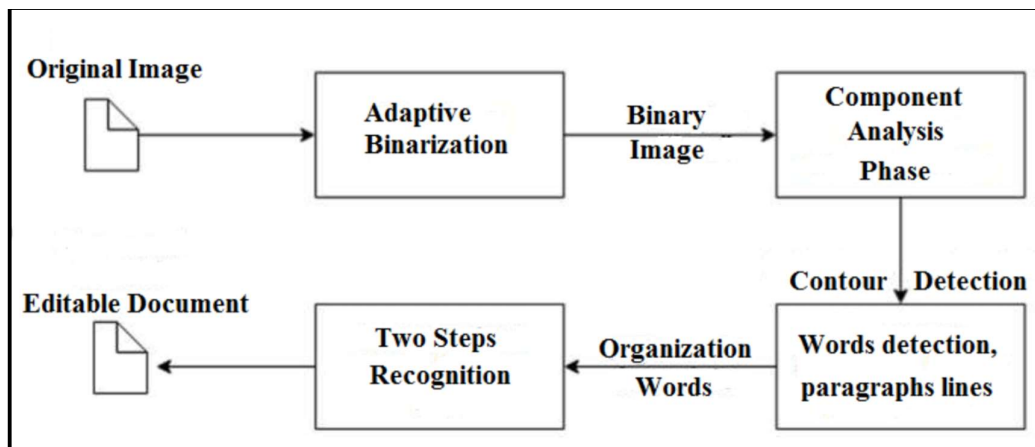


Figure 2.4: Tesseract OCR process from image to text

2.3.2 Limitations in Tesseract:

- Sometimes, OCR is not ideally perfect and right to us as some private enterprise solutions are present to us.
- It is not in the favor of the images that are affected by artifacts including confusing backgrounds.
- It is not capable of recognizing handwriting.
- Average or bad quality OCR can be produced when bad quality image is scanned.

OCR systems can change a typical dual structural image of text that could either consist of machine generated print or might be handwritten text from picture to show into machine readable form. “OCR is widespread use of Neural Network [9]”. A process which contains number of sub divided processes to perform the result accurately is known as OCR. “In the 1950’s OCR was commonly used in data entry applications [10]”.

CHAPTER 3

DESIGN AND METHODOLOGY

3.1 Dataset Collection

The collection of dataset is made through the prescription of local doctors. We are collecting the prescriptions of doctors and labelling the medicine images with the real name in digital text.

The images are captured from 13 mega pixels camera of smartphone, so the quality of image is fine but the size and resolution of images are very huge so we have to follow the following steps in order to prepare the dataset like IAM dataset of handwritten images.

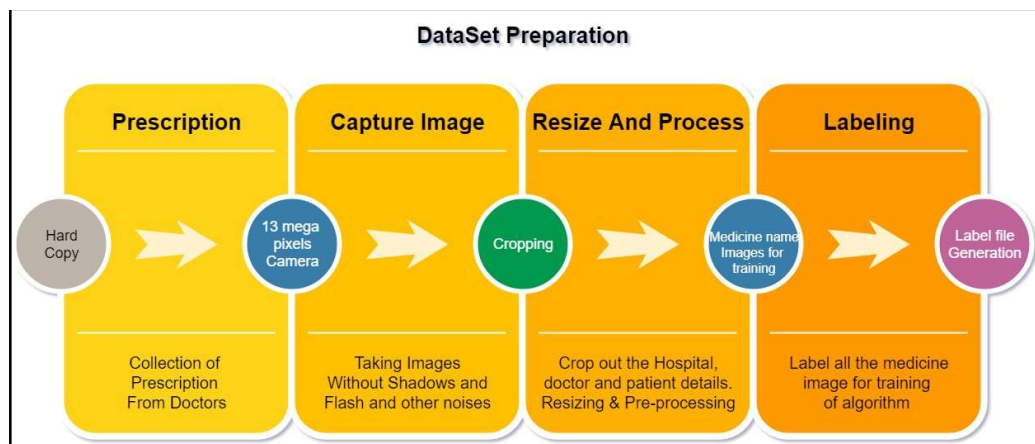


Figure 3.1: Flowchart of dataset Construction

Labelling of images is done through the searching on rxlist.com website and other local searches on Google.

DEA# GB 05455616 LIC # 976269

MEDICAL CENTRE
824 14th Street
New York, NY 91743, USA

NAME John Smith AGE 34
ADDRESS 162 Example St, NY DATE 09-11-12

R_x

Betaloc 100mg - 1 tab BID
Dorzolamidum 10 mg - 1 tab BID
Cimetidine 50 mg - 2 tabs TID
Oxprelol 50mg - 1 tab QD

Dr. Steve Johnson
signature

☐ LABEL
REFILL 0 1 2 3 4 5 PRN

WTS/N-PRESC-T
1-888-422-0760

Figure 3.2: A general & standard medicine prescription

3.2 Labelling of our collected data set

Medicine	Count	Medicine	Count	Medicine	Count
fabrol DS	1	doxy(cap)	11	disprin(tab)	2
iberetfolic	2	pc	7	betneso(eye-drop)	1
CPM(tab)	19	gravinate	1	dyclo(tab)	2
CPM(syrup)	2	di	1	voren(tab)	3
chlorepherinamine	1	omi	1	voren(injection)	2
ascard(tab)	19	beridox	1	histazolin(eye-drop)	2
lipiget(tab)	12	avelox	2	chloroptic(eye-drop)	1
capril	2	cipro	1	ciprin	1
lasix	3	trisol	2	fml(eye-drop)	2
sustac	1	revitale	1	prefrin(eye-drop)	1
concor	2	augmentin(tab)	9	flurozine(eye-drop)	13
zantac(tab)	5	domel(syrup)	1	methachlor(eye-drop)	1
leflor(tab)	4	cal-one-d(tab)	1	pred forte(eye-drop)	2
cefiget	2	vidaylin(syrup)	1	decon-a(eye-drop)	1
flagyl(tab)	12	sangobion(syrup)	1	cefspan(syrup)	1
ORS	4	synflex(tab)	1	rigix	11
motilium(tab)	5	enflor(sachet)	1	regix(tab)	5
peptiban	1	nuberol forte(tab)	1	mytika(scht)	4
urixin(tab)	3	arinac(tab)	1	augmentin(syrup)	2
citrosoda	1	entamizole(tab)	1	acefyll(syrup)	5
vermox	1	novidat(tab)	1	mytika(tab)	2
folic acid	3	citralka(syrup)	1	calpol(syrup)	2
panadol(tab)	19	risek(tab)	1	zincat od(syrup)	3
novidat(syrup)	3	ciproxin	1	enterogermina	3

Table 1.3: Labelling of our dataset

3.3 Working Techniques

The working technique is divided into 5 phases. The 1st phase is image pre-processing of prescription image. The 2nd phase is extracting features from the image which will be feed in to the algorithm. The 3rd phase is dataset loading in to the code for training of the deep learning algorithm of convolutional neural network (CNN). "CNN enables data-driven learning and extraction of highly representative image features from appropriate training data [11]". The 4th phase is training of model with the dataset and in the 5th phase testing of input image with the model of algorithm we have built in the previous phase.

The accuracy of the model can be measured through after comparing the result and labels of test dataset.

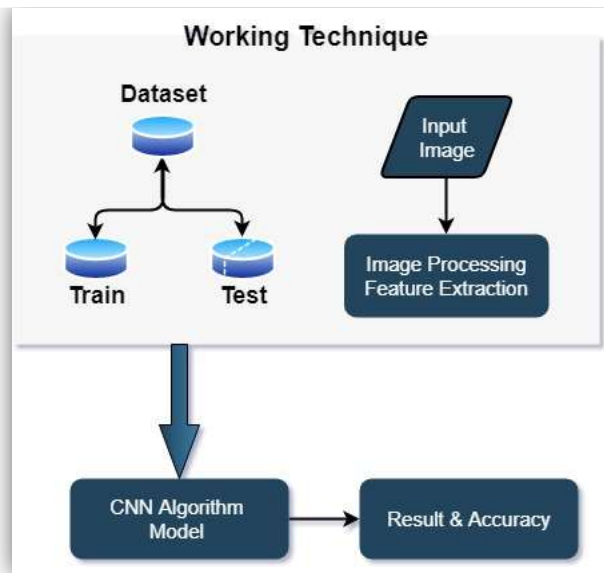


Figure 3.3: Working Technique

3.3.1 Image Pre-processing

“The pre-processing is performed as a binarized image to convert a grayscale image, and noise is reduced on the input image [12]”. It is of no concern that what the image is, it must be without noise that is present due to abasement, visible damages and even those borders from the scanning equipment. Further the image will be gone through a pipeline of steps and algorithms and converted into a clear image with marked lines for the feature extraction phase.

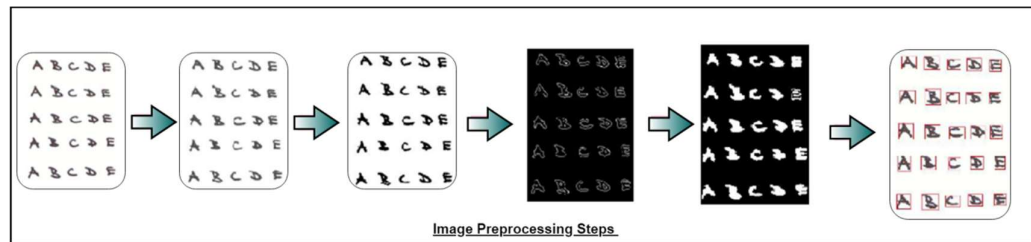


Figure 3.4: Image processing pipeline

The detailed flow chart of image processing with all the steps, loop backs and connections are given below:

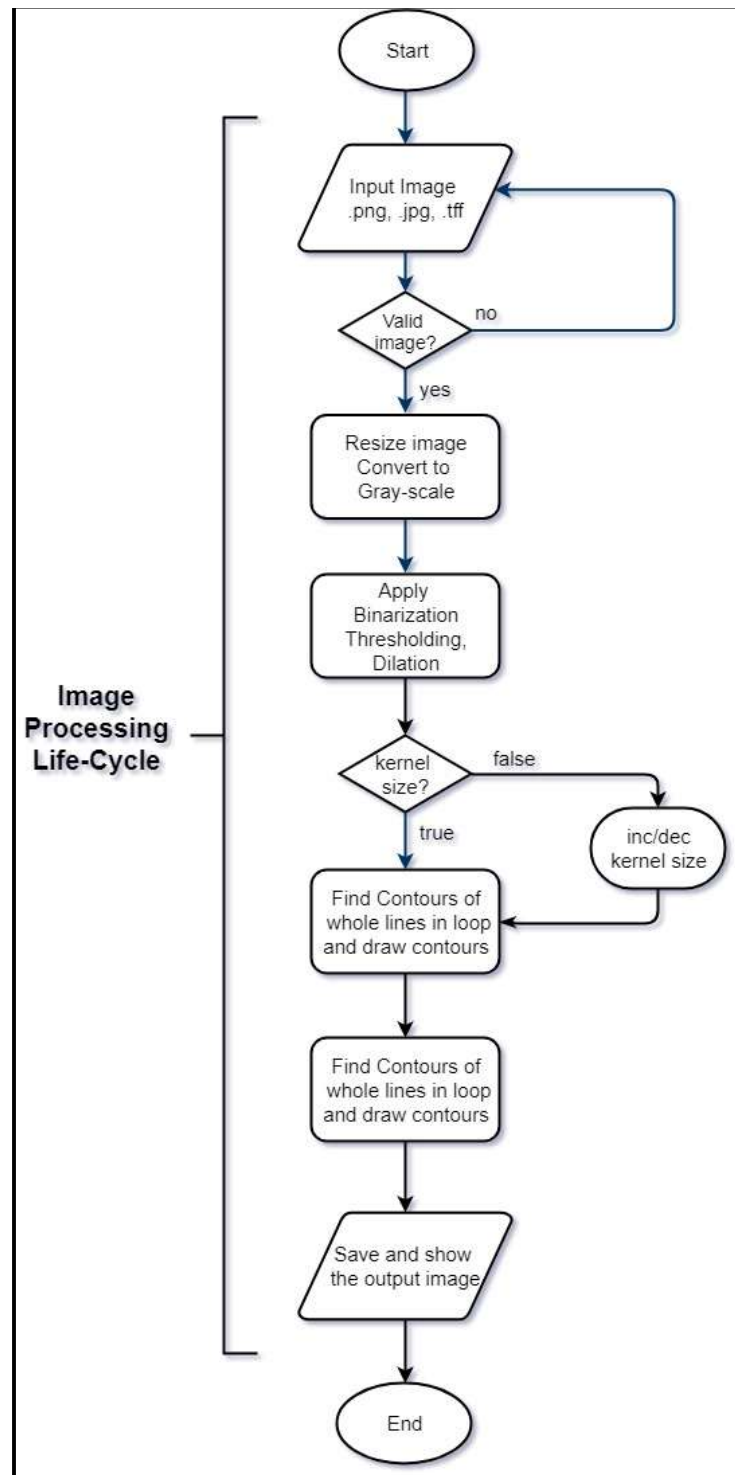


Figure 3.5: Image processing flow chart

3.3.2 Feature Extraction

For making the “Neural Network” understand pattern of doctors’ handwriting, instead of moving each of the sentences or whole lines, just the words of lines are of use. For extracting segment of words from each line, we will segment the lines into words and words into characters. After then dilation and contrast will be set of every word according to the value of pixels. And all the lines, words and characters images will be extract out and save into the directory or folder respectively.

The detailed flow chart of feature extraction with all the necessary steps is given below:

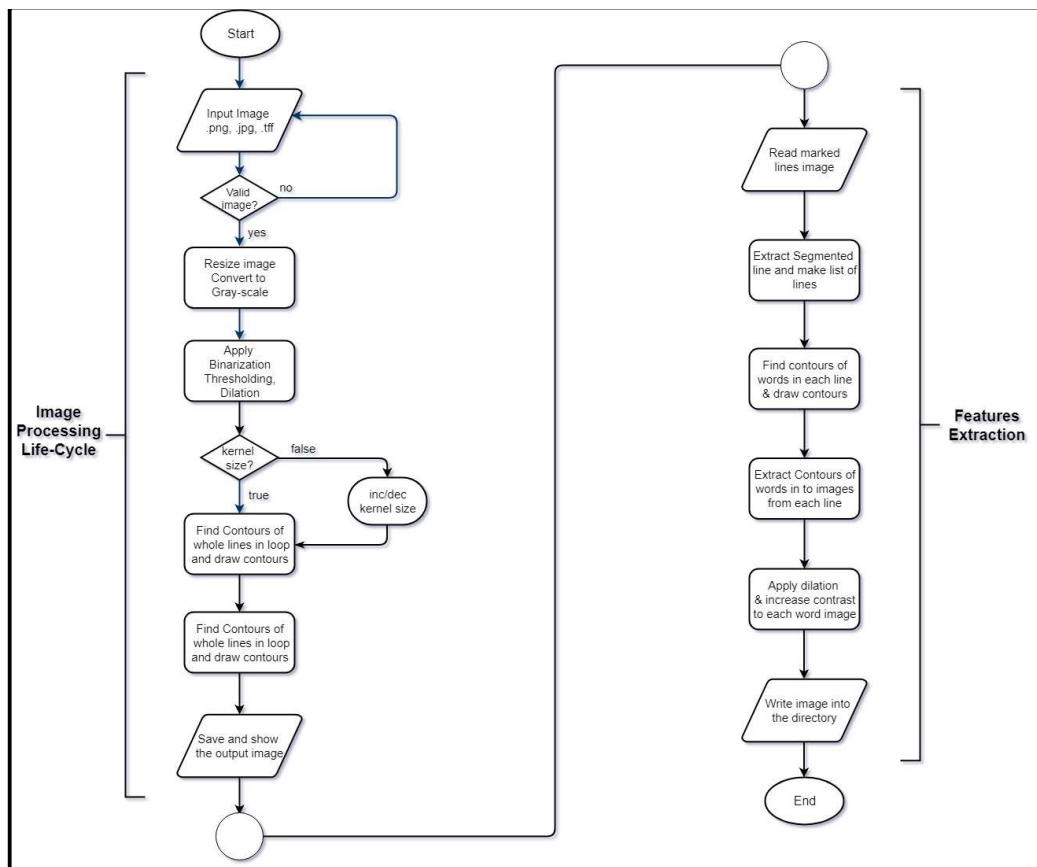


Figure 3.6: Feature extraction flow chart

3.3.3 Character Recognition Algorithm

After segmentation each character separated is feeding into a “Convolutional Neural Network “(CNN).

In the machine learning, to create a model is one of the most essential part. Initially in our model, firstly after collecting dataset & few generator-functions using the python language are formed. In order to implement we would provide with new environment “tensor flow model”. This model is nothing than a building up of CNN in Keras with operating Tensor Flow at back-end.

General Convolutional Neural Network is used with multiple convolutional layers & max pooling layers, with some dense & final-output layers along with “Softmax Activation”. “RELU” and “Softmax activation” functions will be in use among the dense & convolutional layers.

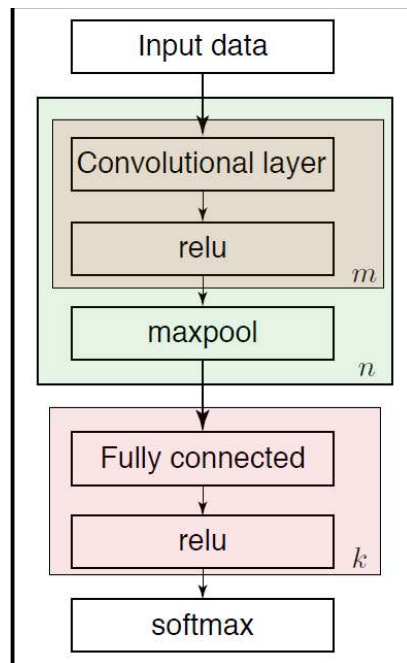


Figure 3.7: CNN layers structure

3.4 Input Output Block Diagram

- **Captured Image:** At this step an image will be taken as an input.
- **Dataset:** A set of finite images which will help in training and testing of model of CNN algorithm
- **Image processing:** The target of using pre-processing is to remove the irregularity that is necessary to be found in cursive handwriting. During this stage of pre-processing, the RGB images are converted to grayscale format and it enhances some image features important for further processing.
- **Feature Extraction:** During this a single words is segmented/split into single character.
- **Recognition:** At this step the trained model of CNN with the train and test set of dataset will feed by the features extracted from input image for the recognition of prescribed medicine.
- **Results:** In this step we find out that our character is recognized or not. If the output needs improvement then we go back again to the feature extraction process.
- **Analysis/Accuracy Measure:** Here we check our accuracy by comparing the recognized character with actual image manually or with the help of confusion matrix.

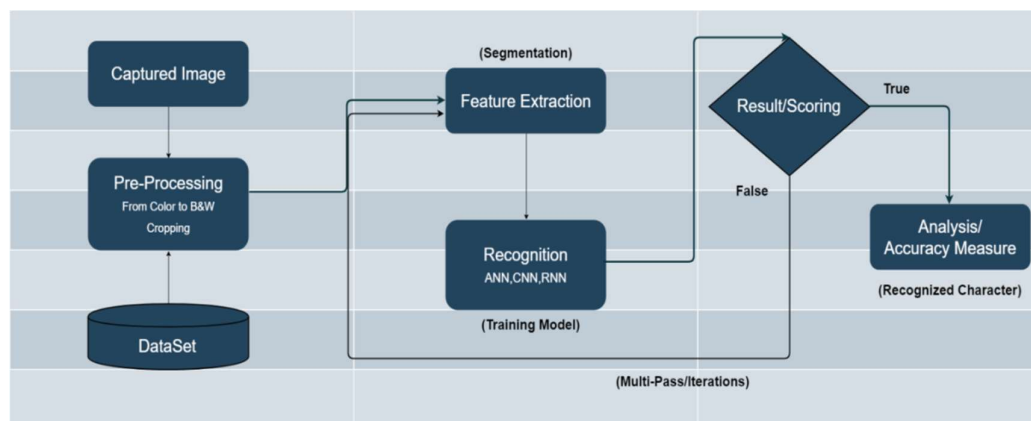


Figure 3.8: IPO flow chart

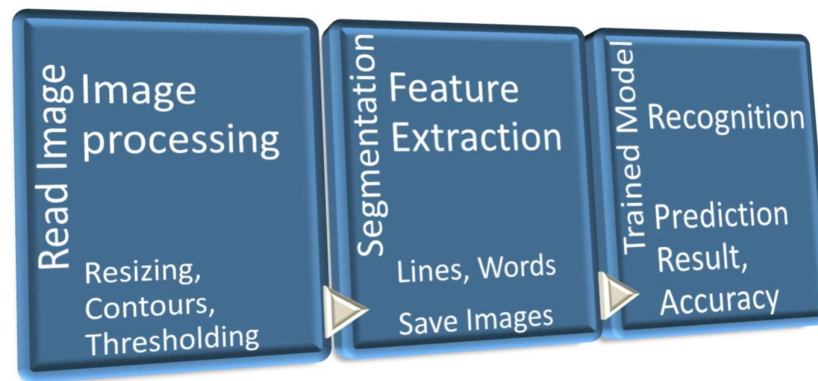


Figure 3.9: IPO Block Diagram

3.5 Implementation & Progress of Image Processing

The first step of image processing is to resize the original size and resolution into a particular standard size of prescription without compromising the quality. The result of original and resize image are shown below:

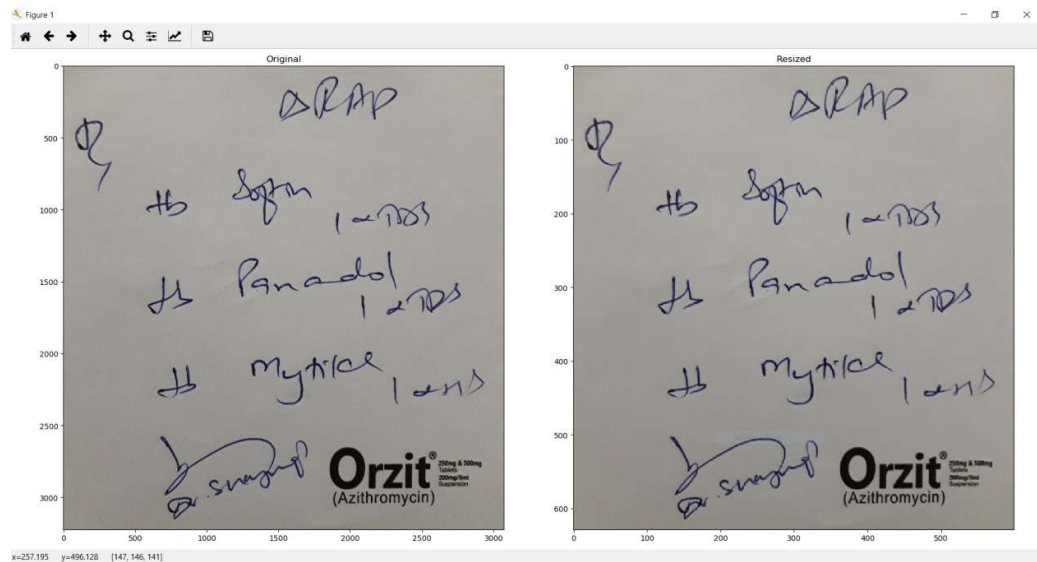


Figure 3.10: Phase 01

The second step is converting RGB into grayscale and into adaptive threshold with contrast and dialation of image and pass the image for feature extraction life cycle.

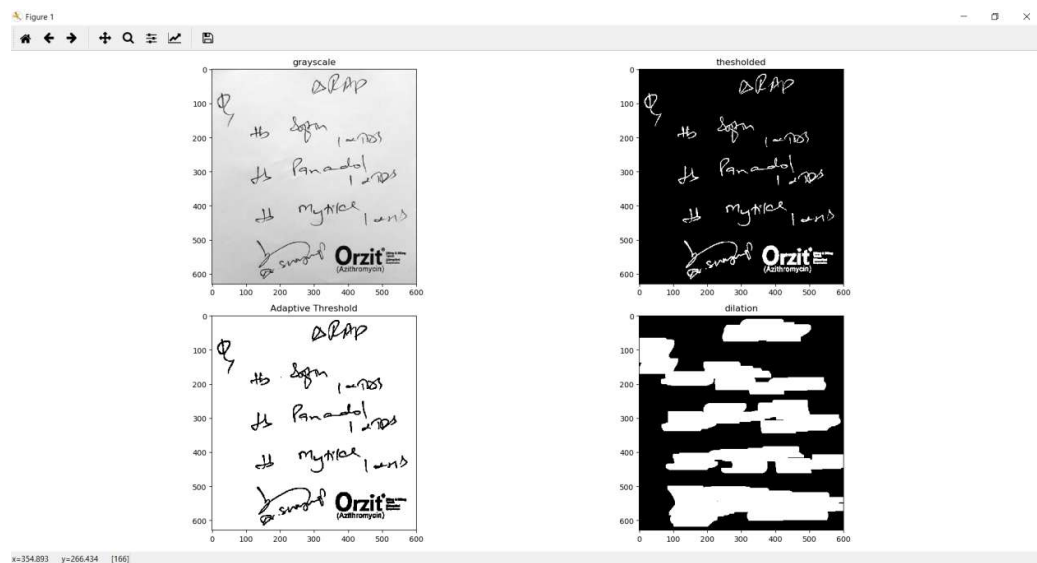


Figure 3.11: Phase 02

The third step is to identify the corners of whole lines in order to draw contours and extract the lines from prescription.

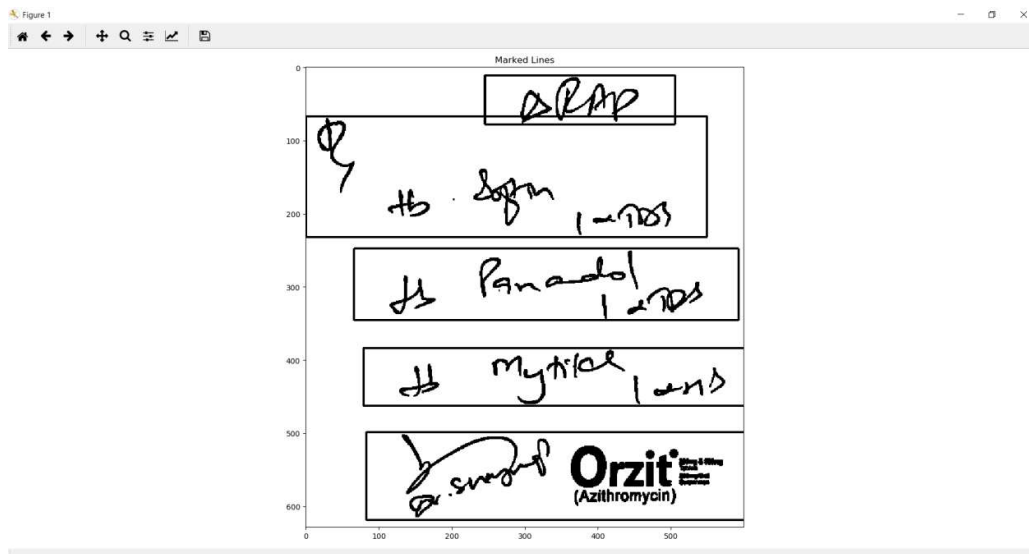


Figure 3.12: Phase 03

The 4th step is to write and save the extracted lines from prescription for word detection and extraction.

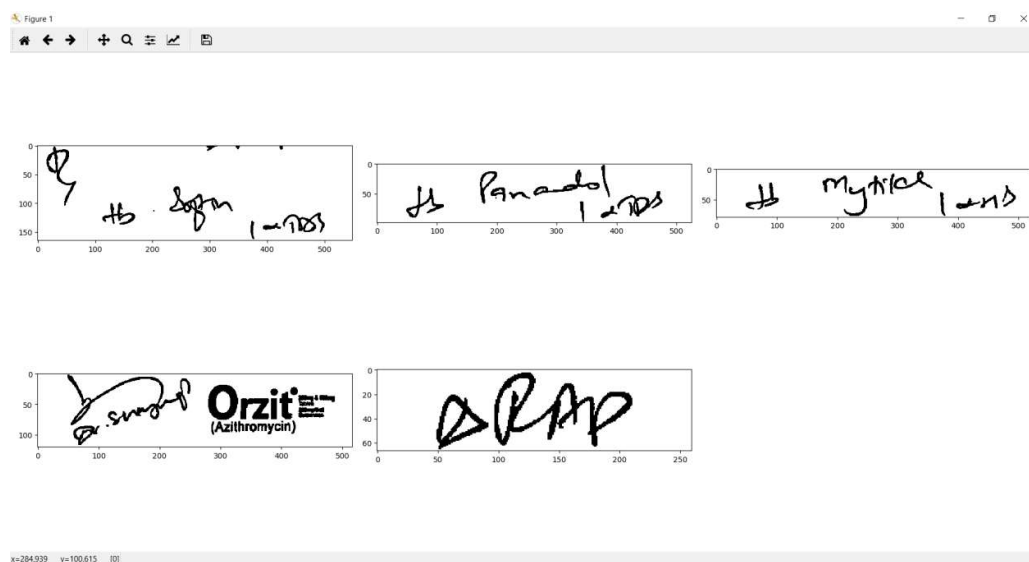


Figure 3.13: Phase 04

The 5th step of image is that extracts the words from lines and detects the corners on the image and draw contours.

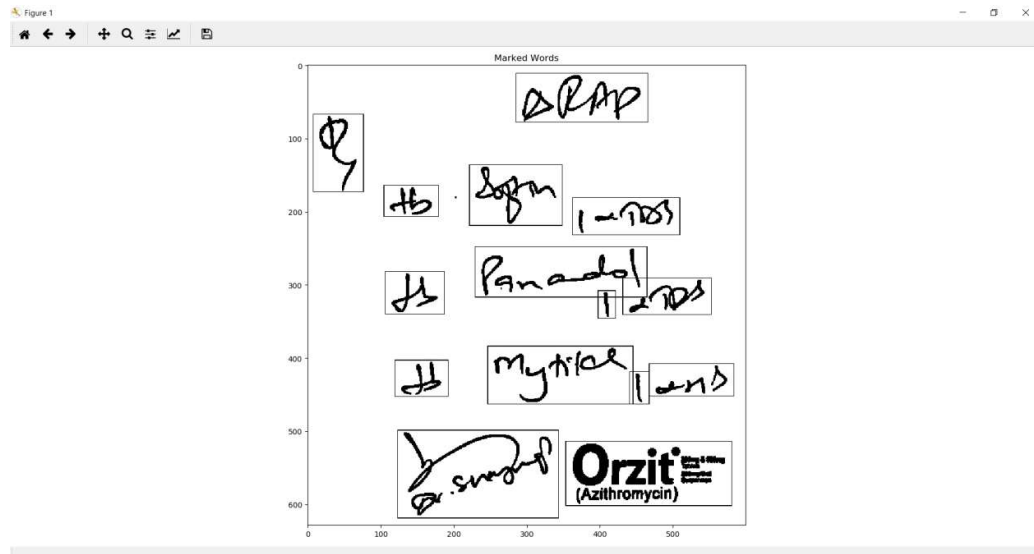


Figure 3.14: Phase 05

The 6th step is to extract the words from image and save it to the directory for further processes of feature extraction life cycle.

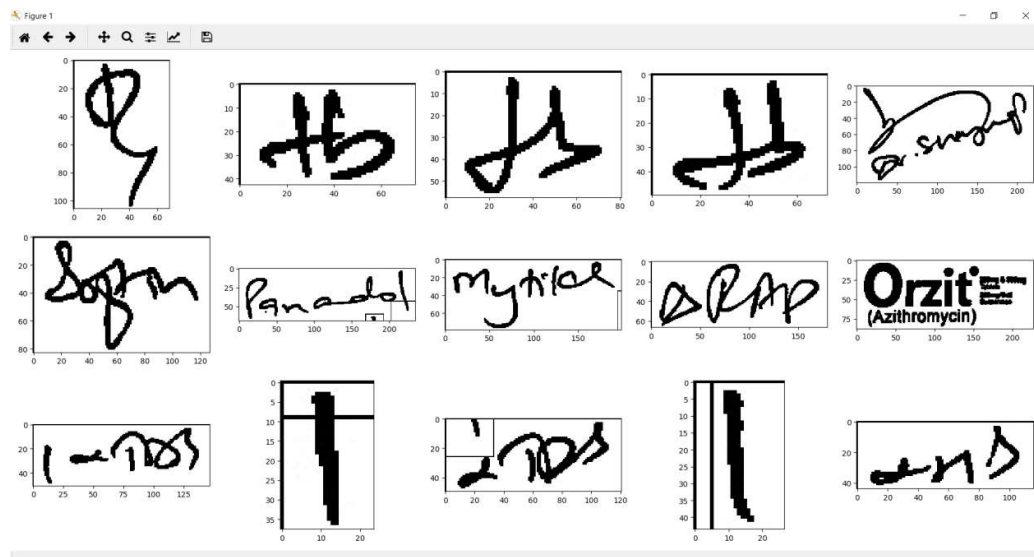


Figure 3.15: Phase 06

CHAPTER 4

IMPLEMENTATION

4.1 Generating Image Dataset

After the successful image processing of prescription images and extraction of features from it we applied this approach for generating the dataset for our algorithm training and testing of handwritten medicines names recognition from prescriptions. We have almost 300 medicine prescription and on each prescription there are average 3 to 4 medicines are prescribed by the doctor.

We implemented the image processing algorithm on each prescription and extract the processed, noise proof and clean black and white images of medicines and save them in “JPG” image file format with a unique six digit id e.g. “026606”.

After successfully generating whole image dataset “X” we also generate the label file which contains the actual output and image id e.g. “001000.jpg augmentin”. The “Prescription_label.txt” contains all the image id’s and labels of the dataset. In this way we have generated all the “X & Y” for training and testing of “CNN” algorithm.



Figure 4.1: Image Dataset “X”



Figure 4.2: Label Text file “Y”

4.2 Data Augmentation

Our self-collected dataset of prescription is relatively not enough for training the algorithm with a good accuracy. So we have implemented the data augmentation algorithm to increase the size of our dataset in order to trained the model with better training and validation accuracy.

4.3 Algorithm Implementation

The algorithm implement required a number of steps and stages to execute training and testing to perform the operation of prediction of prescription medicine image. So from now we are starting to write the source code in object oriented form to implement the algorithm of “Convolutional Neural Network” using our own image and labels dataset and the “CNN” algorithm from “Tensorflow Library framework and Keras API” with a number of different utility libraries and their functions and methods.

4.3.1 Importing Libraries and Framework

The first step is to set the environment, install the required libraries and import them in to the main source code. We have used different types of libraries like some of them for data processing and normalization and some of them for data structuring, storing and encoding and visualization.

4.3.2 Loading Dataset

After the importing libraries the next and most important step is to load the whole dataset of images and labels (X, Y) in to the code and memory for processing and normalization. All The images will read in the dimension of 50x50x1 and in grayscale form. The values of image pixels will be stored in the array and array will be appended for all images and bind with the actual output i.e. the label of that particular image. And also we will split our dataset (X, Y) in to training and testing set by the ratio of 80:20. So it will divide into 4 entities like, training_set_x, training_set_y, testing_set_x and testing_set_y.

4.3.3 Normalization and Encoding

On this stage the dataset is in X, Y arrays so we will normalize the X array in the shape of total number of images into 50x50x1. Because the “Tensorflow CNN” only accepts the normalize array of X and also encoded with normalized array of Y. So we have to use the built-in Tokenizer of “Tensorflow” to encode all the labels in to unique integers so each label will be encoded into a unique integer value. And also save this X, Y arrays into the hard drive in the “pickle” file format for future training of algorithm, so in this way we do not have to follow upper steps of reading, binding, normalization and encoding and can directly load the dataset in memory.

4.3.4 Building and Training CNN Algorithm

We have to define the architecture of the model of “CNN” before training and prediction of the dataset.

4.3.4.1 Setting up Model Architecture

This is the first step of building the model; so we have to define the architecture of the model of “CNN” by creating an empty “Sequential” model and then add multiple numbers of layers of “Convolutional, Maxpooling, BatchNomalization, Flatten, Dropout, Fully Connected or Dense Layers with activation functions of ReLU, and Softmax”.

4.3.4.2 Compiling Model

After adding all the layers and activation functions the model is ready for compilation. In compile function we have used the “sparse_categorical_crossentropy” as loss function, “Adam” as learning rate optimizer and “accuracy” as metrics for evaluating the model.

4.3.5 Model Fitting (Training)

Finally we can fit our compiled model on `training_set_x` and `training_set_y`, with the default batch size of 32 and 12-15 numbers of epochs with the validation data of `testing_set_x` and `testing_set_y`.

4.3.6 Model Evaluation

After the long and time consuming task of fitting we can generate the model summary and evaluate and visualize the accuracy of training and also the accuracy of validation which shows how much better the model has been fit or train on our dataset.

4.4 Prediction from CNN Model

Now the model has been trained and ready to predict the `testing_set_x` and `testing_set_y` portion of dataset. And can compare that how much true and false prediction has been performed by the model of “CNN”.

4.5 GUI & Frontend

Now, after all our python code work is finalized, we need to give our python based application some friendly user interface. The purpose of giving it a user interface is to make things easier for other users and making it easy to provide input to the application and display the output. This will make the application accessible to normal people, which was part of our end goal. To achieve this we had several options and we opted to convert our application in to python Flask Application.

4.5.1 Flask Application

Flask seemed to be the easiest and best option for converting out python code to a web app. It lets us connect the python based backend of the application with HTML, CSS and JavaScript based front end. And apart from that, it lets us deploy our application online so that it can be easily made accessible for any user via a URL.

For front end we have used HTML, Bootstrap & CSS to input an image. We also have provided a JavaScript based crop image application so that users can crop their prescription and target the specific word for the input to the application. For uploading the image python code is used at the backend. Our model then picks the uploaded image from the server and processes it and returns the prediction which is displayed at the frontend again.

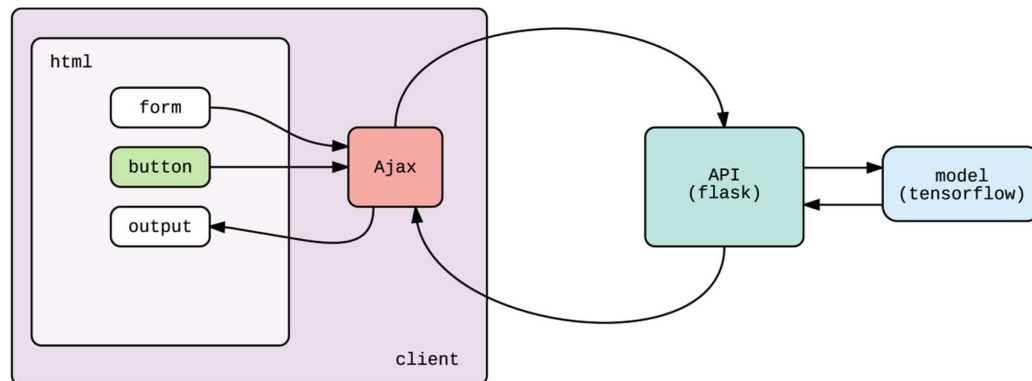


Figure 4.3: Flask Application Process Cycle

4.5.2 Online Deployment

After running and testing the application on local server successfully. We needed to deploy it online so that it can be easily accessible to other people for use. For this we need python supported hosting platform. We can either use free platform i.e “heroku” or we can purchase our own hosting and domain. For now we may opt for heroku which provides great backend support for python as well. It’s free package fits well within our requirements. If the application grows and has strong user base then we can upgrade to paid hosting plans eventually but for now, free hosting will do the job for us.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Result of Prediction

The training accuracy of model is 90+ per cent and the validation accuracy is 70+ per cent. The model has accurately predicted the 70% of the data of testing_set_x. Now this is not the best model but the accuracy is quite better because the size of the dataset is very large but still it can predict the 70% of data accurately.

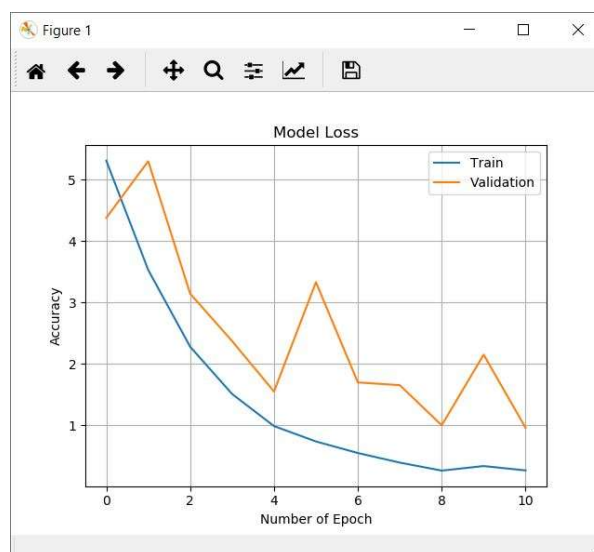


Figure 5.1: Graph of Model Loss

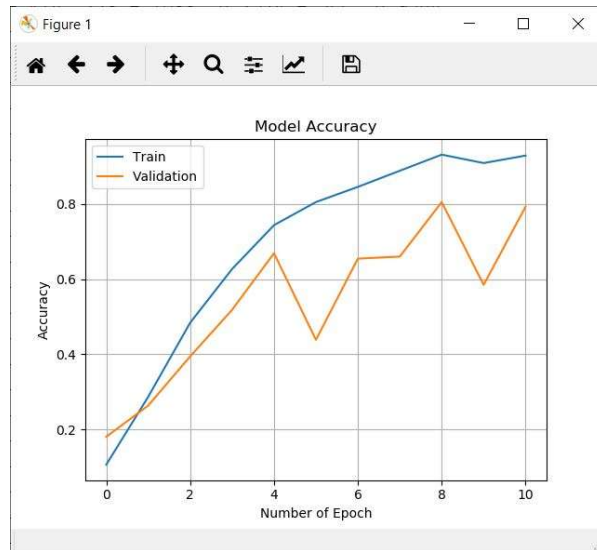


Figure 5.2: Graph of Model Accuracy

The graph shows the increasing in accuracy of training and validation as the number of epochs completed. And the loss in training and validation is decreasing.

5.2 Model Summary

The summary of the models shows all the number of layers and the types of layers on which the model has been trained has and the output shapes from the layers of convolution, batch norm, max pooling , flatten and dense layers.

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 48, 48, 288)	2880
activation_1 (Activation)	(None, 48, 48, 288)	0
max_pooling2d_1 (MaxPooling2D)	(None, 24, 24, 288)	0
batch_normalization_1 (Batch Normalization)	(None, 24, 24, 288)	1152
conv2d_2 (Conv2D)	(None, 22, 22, 288)	746784
activation_2 (Activation)	(None, 22, 22, 288)	0
max_pooling2d_2 (MaxPooling2D)	(None, 11, 11, 288)	0
batch_normalization_2 (Batch Normalization)	(None, 11, 11, 288)	1152
conv2d_3 (Conv2D)	(None, 9, 9, 288)	746784
activation_3 (Activation)	(None, 9, 9, 288)	0
max_pooling2d_3 (MaxPooling2D)	(None, 4, 4, 288)	0
batch_normalization_3 (Batch Normalization)	(None, 4, 4, 288)	1152
dropout_1 (Dropout)	(None, 4, 4, 288)	0
flatten_1 (Flatten)	(None, 4608)	0
dense_1 (Dense)	(None, 288)	1327392
dropout_2 (Dropout)	(None, 288)	0
dense_2 (Dense)	(None, 288)	83232
Total params: 2,910,528		
Trainable params: 2,908,800		
Non-trainable params: 1,728		

Figure 5.3: Summary of Model

5.3 User Friendly and Flexible WebApp

Our flask based application will be keeping the users interacted with the application. The GUI for now is very basic and simple making it very easy to use for the users. We can also convert this web app to android apk file very easily by using android studio' web view feature.

5.4 Comparison between Predicted vs Actual Data

The graph plotting of actual (testing_set_y) and predicted values show comparison between both of data as we can visualize the true and false prediction of the model.

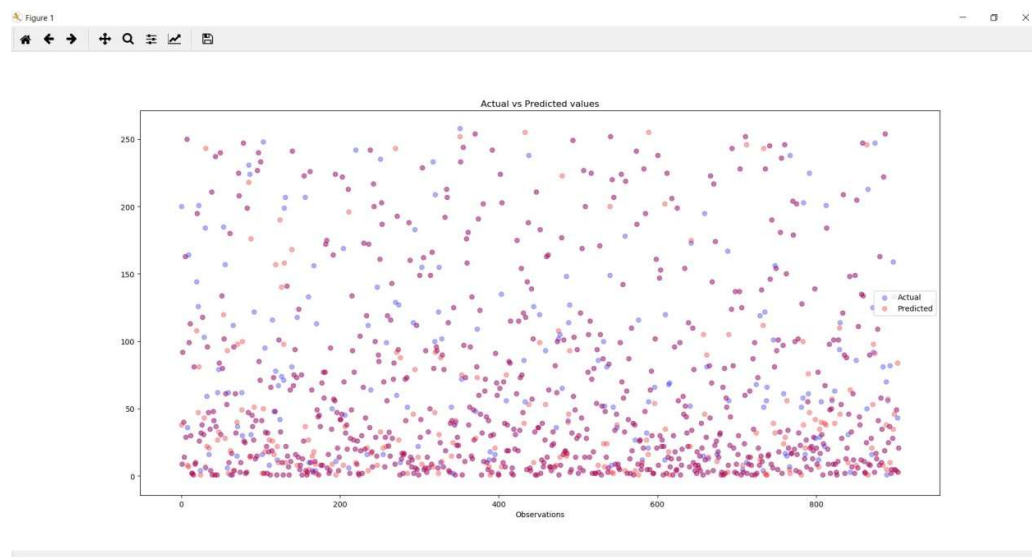


Figure 5.4: Comparison between Actual vs Predicted Values

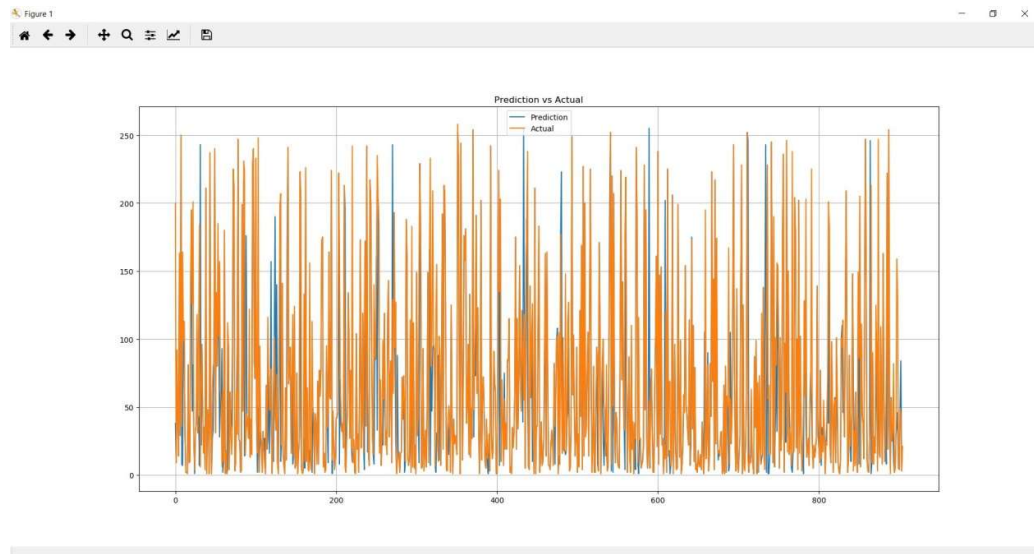


Figure 5.5: Comparison between Actual vs Predicted Values

CHAPTER 6

CONCLUSION AND RECOMMENDATIONS

6.1 Model Accuracy Improvement

Below are the some approaches we thought that can improve the accuracy of model of “CNN”.

6.1.1 Convolutional Layers Parameters

We can try the training of model with different parameters of convolutional layers like adding more layers with different number of filter matrix and the activation function.

6.1.2 More Dropout Layers

The training accuracy is almost 90% and in testing it is 70% in testing which shows it is problem of overfitting of model. This problem of overfitting can be solved through adding more dropout layers or simplify the “CNN architecture” but it is very hard to find the exact number of layers for solving the problem of overfitting.

6.1.3 Data Augmentation

Large number of dataset of different doctors' handwriting can be very helpful for training the model because it will improve the accuracy of prediction and also provide generalized and robustness to the model for predicting different types of doctors' handwritten prescriptions. Also we can do more data augmentation on the dataset to improve the accuracy and prediction.

6.1.4 Improvement in Image Processing

Better image processing of prescription provide less noise, blur, shadows and folding scratches from the prescriptions and the dataset which automatically increases the accuracy and prediction of algorithm.

6.2 Future Scope of Project

Handwriting recognition is presently used in many of the fields of computer science. The technology continues to develop and online information and data exchange still growing day by day. We can see some of the best business fields in which this project can be used for:

6.2.1 Mobile Application of Prescription Automation

The mobile application of this project can be very useful for the ordinary people who cannot understand the complex handwriting of doctors' prescription. So in this way they can easily scan the prescription and predict the medicine name easily in seconds without any obstacles and problem.

6.2.2 Improving the handwriting of Kids

If we train model for normal handwriting of kids and train the model to predicts the written words or letters. So the correct and clean letters or words will be easily identified by the model and the incorrect letters or words will not be predict or identified in this way the kids can improve their handwriting for the incorrect letters which are not identified by the model. This type of system can be very useful for the kids to improve their handwriting and also the generalization of handwriting of all the kids.

6.3 Conclusion

Although our project accurately predicted almost 70% of the test dataset, but there are also some of factors on which we can work to improve further “CNN model”.

The performance of the algorithm can be increased by training the model on large and generalized dataset and optimizing the number of layers, parameters, filters, and depth and activation functions.

Due to lockdown in whole country we are unable to discuss the problems and project in deep with the supervisor and also in the group of ourselves but we have done our best for this project. The project is completed and that's it from our end!

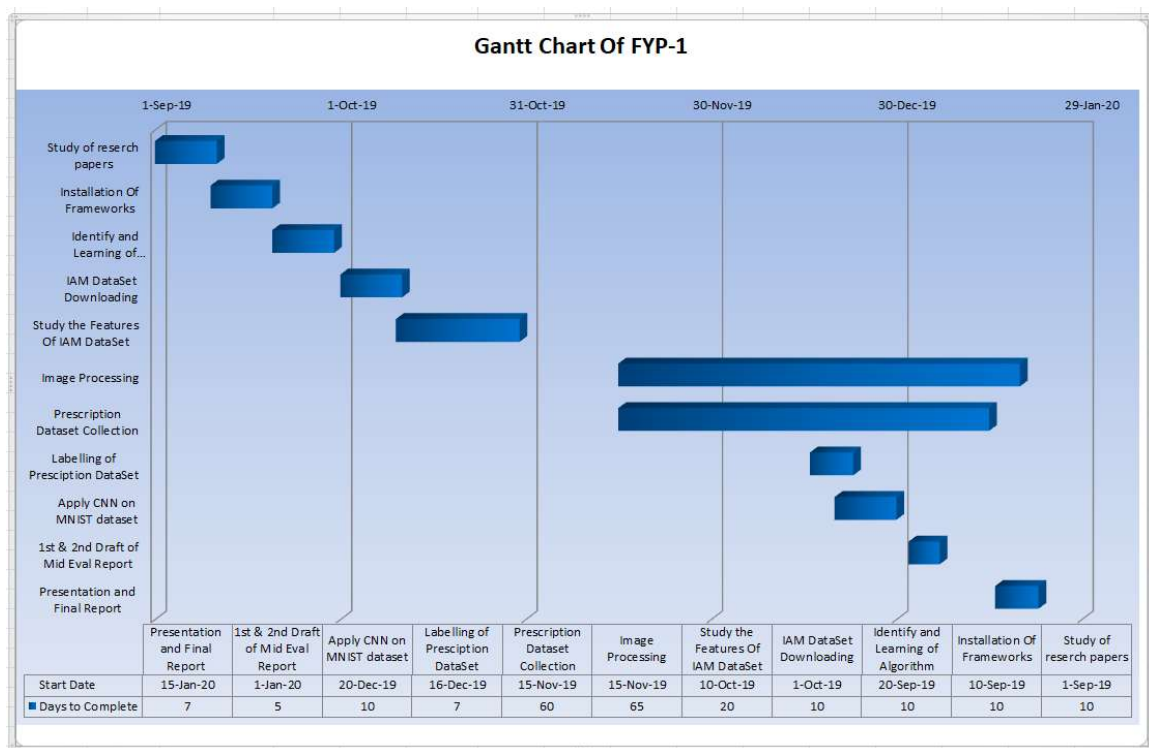
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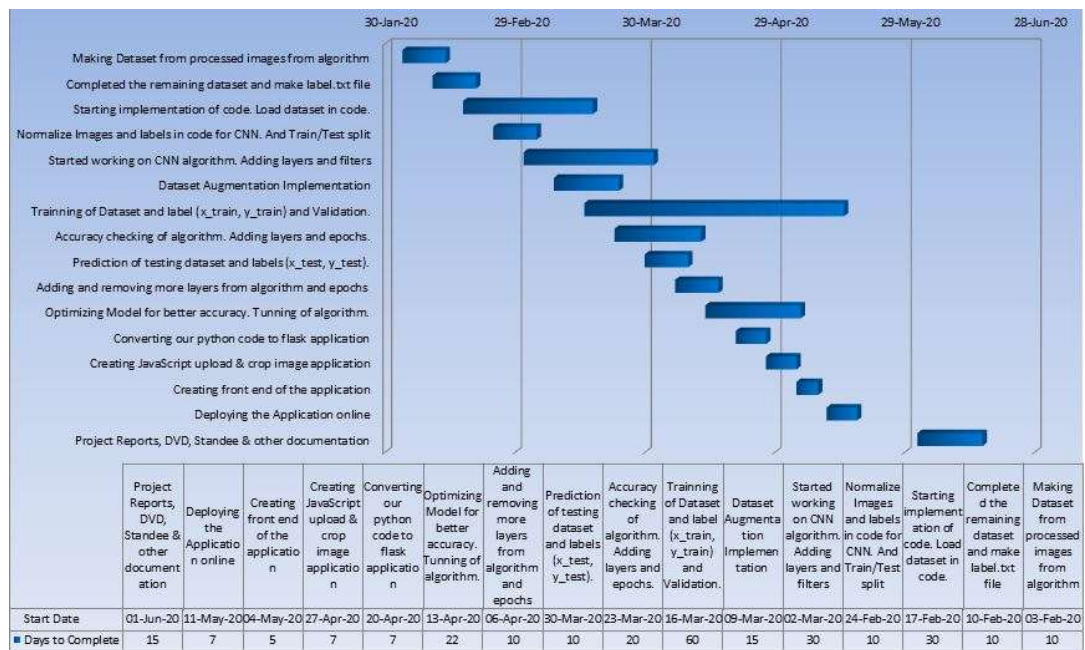
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APPENDICES

APPENDIX A: Gantt chart



Gantt chart FYP-II



APPENDIX B: Dataset Credits

We would like to thank Sindh Government Hospital (New Karachi Branch) for helping us out in our dataset collection. We would also like to thank all the doctors who contributed to this data set. A few of the respected contributors to the dataset are mentioned below:

- Dr. Noor M Khawaja (Sindh Govt. Hospital – DOTS TB Department)
- Dr. Rubina Akhtar (Sindh Govt. Hospital – DOTS TB Department)
- Dr. Iftikhar Memon (Sindh Govt. Hospital – DOTS TB Department)
- Dr. Yousuf (Sindh Govt. Hospital – Chest Medical OPD Department)
- Dr. Farhat (Sindh Govt. Hospital – Chest Medical OPD Department)
- Dr. M. Ali (Sindh Govt. Hospital – Eye Department)
- Dr. Khalid Majeed (Sindh Govt. Hospital – Eye Department)
- Dr. Shagufta (Sindh Govt. Hospital – Children TB Department)
- Dr. Yasir (Sindh Govt. Hospital – Cardiology Department)
- Dr. Shahid Mansa (Sindh Govt. Hospital – Cardiology Department)
- Dr. Shoaib (Sindh Govt. Hospital – Diabetology Department)
- Dr. Rasheed Lakhia (Skin Centre Saddar)
- Dr. Unsa Iqbal (JMPC Karachi)
- Dr. Suman Usmani (DOW University Hospital)
- Dr. Atta Ur Rehman (City Hospital - Saadi Town)